Tanmay Sharma

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Education

Northeastern University **September 2024 – Present**
Master of Science in Cybersecurity - 3.58/4.00 *Boston, MA*

Coursework: Network Security, Linux Administration, Cryptography, Binary Exploitation, Shell Scripting, Security Risk Management and Assessment

SRM Institute of Science and Technology **September 2020 – July 2024**
Bachelor of Technology in Computer Science and Engineering - 3.56/4.00 *Chennai, India*

Coursework: Operating Systems, Computer Architecture, Data Structures, Algorithms, Computer Networks

Experience

CyberspACe securiTy and forensIcs lab (CactiLab) **March 2025 – August 2025**
Research Volunteer *Boston, MA*

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

Institute for Plasma Research (IPR) **September 2023 – November 2023**
Network Systems Trainee - Cybersecurity *Gandhinagar, India*

- Designed and deployed a secure, cost-effective network of Raspberry Pi-based access points, creating a dedicated wireless-to-wireless VPN/Wi-Fi solution for an isolated laboratory of 25–30 plasma physicists, enabling seamless and secure intra-lab communication.
- Implemented advanced encryption and security protocols, reducing network vulnerability by 60% and ensuring compliance with stringent research data protection standards, critical for sensitive plasma research experiments.
- Optimized network performance through real-time monitoring and custom configuration, achieving 99% uptime and improving data accessibility for researchers, which enhanced collaboration and accelerated experimental workflows by 30%.

Projects

Secure Instant Messaging System | *Python, SRP, ECDH, TCP Sockets* **January 2025 – April 2025**

- Built a secure terminal-based chat system using SRP, ECDH, and AES-GCM for confidential, authenticated real-time communication (CY6740 - Network Security).
- Developed socket-based architecture with encrypted commands, rate limiting, and user listing, cutting abuse vectors by 80% and boosting resilience against attacks.

OSINT-Recon LLM Framework | *Python, Mistral 7B, FAISS* **October 2025 – Present**

- Developed an AI-powered OSINT reconnaissance framework integrating Mistral 7B LLM with automated subdomain enumeration, DNS resolution, and HTTP service discovery tools for intelligent security analysis for OSINT phase of PNPT examination.
- Implemented RAG (Retrieval-Augmented Generation) system using FAISS vector database and sentence transformers for semantic search, with asynchronous tool orchestration generating automated threat intelligence reports.

Platform for Azure Response & Security Event Correlation | *Azure, AMA, Azure Sentinel* **November 2024**

- Designed and deployed a cloud-based SIEM using Azure Sentinel, Log Analytics, and a Windows VM to collect and monitor security events in real time.
- Integrated Windows Security Events via Azure Monitor Agent (AMA), enabling centralized log analysis and scalable threat detection across virtual infrastructure.

Certifications, Achievements & Community

Certifications: CompTIA Security+ | AWS Academy Cloud Security Foundations | PNPT by TCM Security (In Progress)
Achievements: SimSpace Cyber Cup 2025 - Rank 6, iCTF Rank 45 out of 1414
Community: Member of Boston Mesh Project and Boston Open Dev Community

Technical Skills

Languages: C/C++, Python, Linux x86 Assembly (32bit/64bit), Bash, PowerShell, DuckyScript, SQL, YAML
Security Tools: Splunk, Azure Sentinel, Wireshark, Nmap, Sysmon, Ghidra, VirusTotal
Technologies: Cryptographic Protocols, SSL/TLS, IPsec, OAuth, Docker, Git, Linux Administration, Git
Frameworks/Standards: MITRE ATT&CK, NIST CSF, ISO 27001, OWASP Top 10, CMMC